

Saturday, 1 December, 2021

Our Workflow

1. Getting familiar with League of Legends

In this step, we have been trying to understand the game. It is a strategic game with so many variables at place. So, our first step was understanding what each role means, how a champion can improve, the in-place obstacles each team has to overcome, and understanding items and build and how an item interacts with a champ's keystone.

2. Getting familiar with Riot API to begin data collection

We got familiar with how the Riot API works and what kind of information we can get. We can get:

- Service status.
- Summoner details: name, level, profile icon, account ids.
- Match History and Match details and timelines by Summoner name.
- Someone's current game info, if someone is in the game.
- Other information

Riot API can also provide access to live client data and replay data. This can be used to build a basic live client script to give information about games online and in real time.

Data Analysis

1. Champions win, pick, and ban rates

We will get the champions names from the participants of each match; 10 champs per match. Win status can be used to get the status of win/lost for each champion we have. Then, using the sum of all matches a champion played, we can get the win rate.

For pick rate, we would get the total matches played by each champion. Then, we would divide it by the total number of matches.

For bans. We'd get them from the 'bans' struct. Then, calculate the ban rate for each champ.

2. Champions Synergies or Duos

We'll probably use the data from the first requirement to get the duos loss or win rate. Then, we would calculate the synergy factor.

3. Item Win, Pick Rates

There is a win status column that can be used to get the win count for each item. As for the pick count, we would need to be cautious of how an item, unlike a champion, can be picked multiple times a game by different participants. For both this and the first requirement, we would visualize the highest and lowest win, pick rates.

4. Item Synergies

Again, we would use data from previous requirement to get once the champion and once with the general class. Then, a calculation for synergy is performed.

5. Item Suggestion

We are still considering multiple approaches to this. Will it be on-line or not? And what factors should we consider other than previous win rates of similar match?

6. Creative Requirements

We are thinking:

- ❖ Lane suggestion based on your champion and your teammates' champions.
- ❖ Worst Enemy : which champions are your champion most vulnerable against.